

Michael J. Sieler Jr., Ph.D.

Department of Genetics and Evolution
University of Geneva
Quai Ernest-Ansermet 30, 1205 Geneva, Switzerland

Michael.SielerJr@UniGe.ch
[MichaelSieler.com](https://michaelsieler.com)
GitHub: [sielerjm](https://github.com/sielerjm)
LinkedIn: [mjsielerjr](https://www.linkedin.com/in/mjsielerjr)

EDUCATION

Oregon State University, Corvallis, OR Sept 2025
Ph.D. Microbiology. GPA: 3.95
Dissertation: "Defining the context dependence of how the zebrafish (*Danio rerio*) gut microbiome responds to environmental stressors"

Oregon State University, Corvallis, OR June 2020
B.S. Bioresource Research, options: Bioinformatics and Genomics. GPA: 3.82

RESEARCH APPOINTMENTS

Postdoctoral Researcher, Liberti Lab 2026-Present
Research focus in gut microbiota-brain-axis of eusocial bee model systems
Department of Genetics and Evolution, University of Geneva

Postdoctoral Researcher, Sharpton Lab 2025
Microbiome data science
Department of Microbiology, Oregon State University

Ph.D. Graduate Student Researcher, Mentor: Thomas J. Sharpton 2020-2025
Research focus in host-associated microbiome stability to environmental stressors
Department of Microbiology, Oregon State University

Ph.D. Intern, Mentor: Lisa Bramer and Kelly Stratton 2023-2024
Research focus in metabolomic data science and bioinformatics
Pacific Northwest National Laboratory

Undergraduate Student Researcher, Mentor: Thomas J. Sharpton 2018-2020
Research focus in zebrafish microbiome ecology and bioinformatics
Department of Microbiology, Oregon State University

Undergraduate Student Researcher, Mentor: Taifo Mahmud 2017-2018
Research focus in identifying novel antibiotic compounds
Department of Pharmacy, Oregon State University

SCIENTIFIC PUBLICATIONS

Michael J. Sieler Jr., Colleen E Al-Samarrie, Kristin D Kasschau, Michael L Kent, Thomas J Sharpton (2025). "Modelling the zebrafish gut microbiome's resistance and sensitivity to climate change and infection" [Front. Microbiomes](#).

- Designed and conducted experiment; collected 260 fecal samples; conducted gut microbiome statistical analyses; contributed to the writing and editing of the manuscript; and prepared the figures.
- GitHub Repo:** https://github.com/sielerjm/Sieler2025_ZF_Temperature_Parasite

Scientific publications continued on next page.

Keaton Stagaman, Alexandra Alexiev, **Michael J. Sieler Jr.**, Austin Hammer, Kristin D. Kasschau, Lisa Truong, Robyn L. Tanguay & Thomas J. Sharpton (2024). "The zebrafish gut microbiome influences benzo[a]pyrene developmental neurobehavioral toxicity". [Sci Rep](#).

- Collected >1000 zebrafish embryos, exposed them to treatments (including germ-free derivation) and plated them; dissected >100 intestines and prepared intestines for DNA extraction; and contributed to the writing and editing of the manuscript.

Austin Hammer, Christopher Gaulke, Manuel Garcia-Jaramillo, Connor Leong, **Michael J. Sieler Jr.**, Jeff Morré, Yuan Jiang, Claudia Maier, Michael Kent, Thomas Sharpton, and Jan Fred Stevens (2024). "Gut microbiota metabolically mediate intestinal helminth infection in Zebrafish". [mSystems](#).

- Designed figure displaying experimental design schematic.

Michael J. Sieler Jr., Colleen E Al-Samarrie, Kristin D Kasschau, Zoltan M Varga, Michael L Kent, Thomas J Sharpton (2023). "Disentangling the link between zebrafish diet, the gut microbiome succession and *Mycobacterium chelonae* infection." [Anim. Microbiome](#).

- Conducted gut microbiome statistical analyses; contributed to the writing and editing of the manuscript; and prepared the figures.

Joseph A Szule, Lawrence R Curtis, Thomas J Sharpton, Christiane V Löhr, Susanne M Brander, Stacey L Harper, Jamie M Pennington, Sara J Hutton, **Michael J. Sieler Jr.**, Kristin D Kasschau (2022). "Early Enteric and Hepatic Responses to Ingestion of Polystyrene Nanospheres from Water in C57BL/6 Mice." [Front. Water](#).

- Performed the gut microbiome and integrated statistical analyses; contributed to the preparation and editing of the manuscript; and prepared the figures.

Maude M David, Christine Tataru, Quintin Pope, Lydia J Baker, Mary K English, Hannah E Epstein, Austin Hammer, Michael Kent, **Michael J. Sieler Jr.**, Ryan S Mueller, Thomas J Sharpton, Fiona Tomas, Rebecca Vega Thurber, Xiaoli Z Fern (2022). "Revealing General Patterns of Microbiomes That Transcend Systems: Potential and Challenges of Deep Transfer Learning." [mSystems](#).

- Contributed to writing and editing the main manuscript text.

Thomas J. Sharpton, Keaton Stagaman, **Michael J. Sieler Jr.**, Holly K. Arnold, Edward W. Davis II (2021). "Phylogenetic integration reveals the zebrafish core microbiome and its sensitivity to environmental exposures." [Toxics](#).

- Contributed to data curation, writing and editing the main manuscript text.

IN-PREP & UNDER REVIEW PUBLICATIONS

Gregorio Calderoni, **Michael J. Sieler Jr.**, Enrico Giammatteo, Chrystelle Perruchoud, Joanito Liberti. "Using fecal samples as a reliable proxy for bumblebee (*Bombus terrestris*) gut microbiota community analysis". *In-prep*.

- Processed microbiome samples and analyzed microbiome data; contributed to the writing and editing of the manuscript; and prepared final manuscript figures.

Michael J. Sieler Jr., Connor Leong, Kristin D Kasschau, Thomas J Sharpton. "System-Agnostic Dynamical Microbiome Measures (SADMMs) for longitudinal multi-study and -system microbiome analysis". *In-prep*.

- Conceptual development of metrics; contributed to the writing and editing of the main manuscript and figures.

Michael J. Sieler Jr., Connor Leong, Kristin D Kasschau, Michael L Kent, Thomas J Sharpton. "Evaluating the effects of stressor history on zebrafish gut microbiome's resistance and resiliency to parasite exposure". *In-prep*.

- Designed and conducted experiment; designed and conducted experiment; collected >1000 of fecal and intestinal samples for microbiome and transcriptomic analysis; conducted multi-omic statistical analyses; contributed to the writing and editing of the manuscript; and figures.

Damon T. Leach, **Michael J. Sieler Jr.**, Kelly G. Stratton, Rachel E. Richardson, Jennifer E. Kyle, Young-Mo Kim, Josie G. Eder, Kristin M. Engbrecht, Athena A. Schepmoes, Bobbie-Jo M. Webb-Robertson, Lisa Bramer. "Analyzing batch effect correction algorithms for small molecule data using ground truth from a designed experiment." *In-prep*.

- Bioinformatically processed and analyzed 200+ lipidomic samples; contributed to statistical analyses, writing and editing the main manuscript text.

Scientific publications continued on next page.

Emilee Lance, **Michael J. Sieler Jr.**, Colleen E Al-Samarrie, Kristin D Kasschau, Michael L Kent, Thomas J Sharpton. "Investigating the interaction of host genetics and parasite burden on the microbiome in zebrafish". *In-prep*.

- Contributed to statistical analyses, writing, and editing of the manuscript

PRESENTATIONS

Invited Seminar, Department of Genetics and Evolution 2025
University of Geneva *Geneva, Switzerland*

"Defining the context dependence of how the zebrafish gut microbiome responds to environmental stressors"

Invited Seminar, Institute of Molecular Biology 2025
University of Geneva *Eugene, Oregon*

"Defining the context dependence of how the zebrafish gut microbiome responds to environmental stressors"

Connecting Microbiome Communities 2024
Society for Industrial Microbiology and Biotechnology *San Diego, California*

"Modelling the Gut Microbiome's Resistance and Resilience to Climate Change and Infection in Zebrafish"

9th Conference on Beneficial Microbes 2024
University of Wisconsin *Madison, Wisconsin*

"Modelling the Gut Microbiome's Resistance and Resilience to Climate Change and Infection in Zebrafish"

5th Annual MANA Conference 2023
Metabolomics Association of North America (MANA) *Columbia, Missouri*

"Choice of batch correction method is an important factor in small molecule study"

Zebrafish Husbandry Workshop 2022
Aquaculture *San Diego, CA (virtual)*

"Effects of diet on growth and the microbiome"

3rd International Fish Microbiota Workshop 2021
Chinese Academy of Agriculture Sciences *Beijing, China (virtual)*

"Zebrafish laboratory diets differentially alter gut microbiota composition"

PANELS

9th Conference on Beneficial Microbes – Panelist 2024
University of Wisconsin *Madison, Wisconsin*

"The Importance of Inclusive Practices in Microbiome Science"

WORKSHOPS

Microbiome Data Analytics Boot Camp – Trainer 2025
Skills for Health and Research Professionals (SHARP), Columbia University *Virtual (remote)*

"Planning, generating, and analyzing 16S rRNA gene sequencing surveys"

- Supported workshop participants and lecturers in microbiome data analysis and pedagogy

Connecting Microbiome Communities – Trainer 2024
Society for Industrial Microbiology and Biotechnology *San Diego, California*

"Microbiome Metadata Mastery and Research Training: Equipping the Next Generation of Researchers Across Academia, Government, and Industry"

- Co-led workshop, prepared training and demonstration materials

Microbiology Department – Trainer 2024
Oregon State University *Corvallis, OR*

"NMDC: Metadata Standards and Submission Portal for Multi-Omic Analysis"

- Led workshop, prepared training and demonstration
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POSTERS

Biology26 Conference

University of Neuchâtel

2026

Neuchâtel, Switzerland

"Ecological, Developmental, and Social Determinants of Microbial Transmission in the Semelparous Eusocial Bee *Bombus terrestris*"

College of Science Graduate Science Research Showcase

Oregon State University

2025

Corvallis, OR

"Modelling the Gut Microbiome's Resistance and Resilience to Climate Change and Infection in Zebrafish" *Posters*

80th Annual OPHA Conference

Oregon Public Health Association

2024

Corvallis, OR

"The Human Gut Microbiome at the Intersection of Public Health and Social Equity"

ARCS Annual Scholar Event

ARCS Foundation

2022

Portland, OR

"How do external environmental factors impact the gut microbiome to influence host health?"

2nd International Fish Microbiota Workshop

University of Oregon

2019

Eugene, OR

"The gut microbiome drives Benzo(a)pyrene's impact on zebrafish behavioral development"

College of Agriculture Science Showcase

Oregon State University

2019

Corvallis, OR

"The gut microbiome drives Benzo(a)pyrene's impact on zebrafish behavioral development"

TEACHING APPOINTMENTS & MENTORSHIP

Postdoctoral Fellow

Microbiome Analysis (Co-instructor) - University of Geneva

2026

Graduate Teaching Assistant

General Microbiology Lab (Teaching Assistant; MB 303; Spring) - Oregon State University

2022-2023

Human Microbiome (Teaching Assistant; MB 436; Spring) - Oregon State University

2021

Introduction to Microbiology (Teaching Assistant; MB 230, Spring) - Oregon State University

2021

Mentorship

Genetics and Evolution Ph.D. and master's students - University of Geneva

2026-Present

Microbiology Ph.D. and undergraduate students - Oregon State University

2023-2025

FELLOWSHIPS & AWARDS

OSU Scholarly Presentation Award (\$600) <i>Oregon State University</i> Competitive funding to support graduate students presenting their research at professional conferences.	2024
NMDC Ambassador (\$1,000) <i>National Microbiome Data Collaborative</i> Recognized and received training for early career contributions for promoting and leading workshops on findable, accessible, interoperable and reusable microbiome research data and workflows.	2024 – 2025
ODFW Fish Health Graduate Research Fellowship (\$56,000) <i>Oregon Department of Fish and Wildlife</i> Recognized for research in Microbiology at Oregon State University, focusing on fish health issues to benefit Oregon's fish populations.	2023 – 2025
Science Communication Fellow (\$1000) <i>Oregon Museum of Science and Industry</i> Received certified training in informal science education and engagement with public audiences	2020 – 2022
ARCS Scholar (\$18,000) <i>ARCS Foundation</i> Recognized for early significant contributions to scientific research	2020 – 2023

Peer Review

Hansson, B. et al. 2026. "Microbial signaling coordinates group oviposition and predator defense in migratory locusts"

PROFESSIONAL AFFILIATION & SERVICE

Leidholdt Microbiology Summer Camp – Camp Mentor <i>Department of Microbiology, Oregon State University</i> Supervised 20 high school students from historically underrepresented backgrounds in learning laboratory techniques.	2022, 2024, 2025
Food and Nutrition Special Interest Section – Founding Section Member Founding Section Member <i>Oregon Public Health Association</i>	2023-2025 <i>Portland, OR</i>
Microbes and Social Equity Working Group – Member	2022-Present
Microbiology Graduate Student Association – President President <i>Oregon State University</i>	2022 – 2023 <i>Corvallis, OR</i>

ADDITIONAL SKILLS & TRAINING

- **Programming Languages:** R, Python, C# (Unity), Git, HTML, CSS, C++, UNIX/LINUX
 - **Statistics and Data Analytics:** machine learning, multivariate regression, model building and selection, data visualization (Ggplot, plotly, R Shiny)
 - **Bioinformatics:** Nf-core pipelines, 16S sequencing, transcriptomics, metagenomics, metabolomics, phylogenetics, batch effect correction algorithms
 - **Molecular Biology:** zebrafish husbandry, DNA extraction, PCR, gel electrophoresis
 - **Other:** Microsoft Office Suite, Adobe Suite
 - **Languages:** English (Native), German (B2), Spanish (A1)
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